

BANK ACCOUNT MANAGEMENT SYSTEM

Java OOP Laboratory Record

1. Objective

To develop a Java-based Bank Account Management System using classes, objects, inheritance, constructors, method overriding and basic banking operations such as deposit, withdrawal, transfer and interest calculation.

2. Task Description

The program creates a base Account class containing account number, balance and account type. It provides deposit, withdrawal, transfer and account-detail operations. SavingsAccount extends Account and calculates interest. CurrentAccount extends Account and overrides withdrawal to support an overdraft limit. The Bank class contains main() and demonstrates all operations.

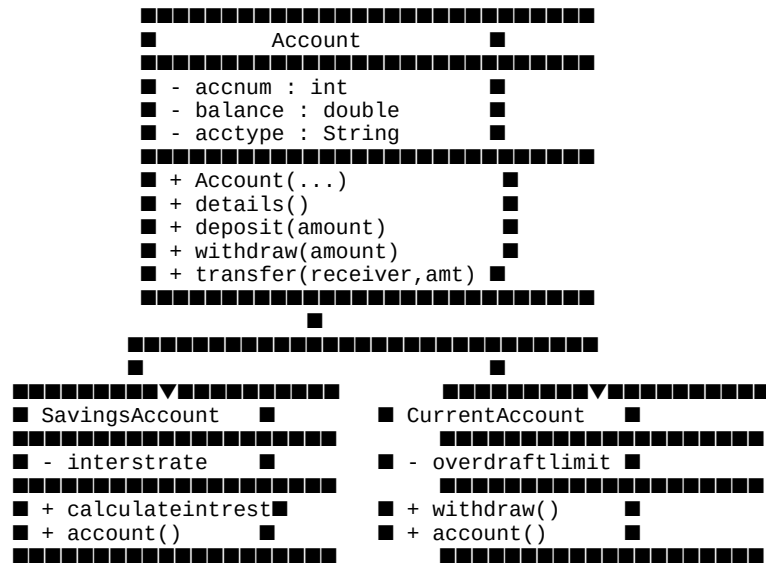
3. Step-Wise Execution

1. Create a SavingsAccount object with account number 2000, balance 2424 and interest rate 5%.
2. Create a CurrentAccount object with account number 2154, balance 5845 and overdraft limit 500.
3. Display the initial details of both accounts.
4. Deposit 1000 into the savings account.
5. Withdraw 2000 from the current account.
6. Calculate 5% interest on the updated savings balance.
7. Transfer 500 from the savings account to the current account.
8. Withdraw 5000 from the current account using its overdraft facility.
9. Display the final account details.

4. Algorithm

1. Start.
2. Create SavingsAccount and CurrentAccount objects.
3. Display their initial details.
4. Perform deposit operation on the savings account.
5. Perform withdrawal operation on the current account.
6. Calculate savings-account interest and add it to the balance.
7. Transfer money from savings account to current account.
8. Perform another withdrawal from the current account using the overdraft limit.
9. Display final balances and account information.
10. Stop.

5. UML Diagram



6. Program Code

```
class Account{
    int accnum;
    double balance;
    String acctype;

    Account(int n,double m,String l){
        this.accnum=n;
        this.balance=m;
        this.acctype=l;
    }

    void details() {
        System.out.println("accnum :"+accnum);
        System.out.println("balance :"+balance);
        System.out.println("acctype :"+acctype);
    }

    void deposit(double amount) {
        if (amount > 0) {
            balance = balance + amount;
            System.out.println("Deposited: " + amount);
        } else {
            System.out.println("Invalid deposit amount");
        }
    }

    void withdraw(double amount){
        if (amount > 0 && amount <= balance) {
            balance = balance - amount;
            System.out.println("Withdrawn: " + amount);
        } else {
            System.out.println("Insufficient balance");
        }
    }

    void transfer(Account receiver,double amount) {
        if(amount > 0 && amount <= balance) {
            this.balance = this.balance - amount;
            receiver.balance = receiver.balance + amount;
            System.out.println("Transferred: " + amount);
            System.out.println("To Account: " + receiver.accnum);
        } else {
            System.out.println("Transfer failed");
        }
    }
}

class SavingsAccount extends Account{
    double interstrate;

    SavingsAccount(int n,double m,double k){
        super(n,m,"saving account");
        this.interstrate=k;
    }

    void calculateintrest() {
        double intrest = balance * (interstrate / 100);
        balance = balance + intrest;
        System.out.println("value :"+intrest);
    }

    void account() {
        System.out.println("accnum :"+accnum);
        System.out.println("balance :"+balance);
        System.out.println("interstrate :"+interstrate);
    }
}

class CurrentAccount extends Account{
    double overdraftlimit;

    CurrentAccount(int n,double m,double o){
        super(n,m,"current account");
        this.overdraftlimit=o;
    }

    @Override
    void withdraw(double amount){
        if (amount > 0 && amount <= balance+overdraftlimit) {
            balance = balance - amount;
            System.out.println("Withdrawn: " + amount);
        }
    }
}
```

```

        } else {
            System.out.println("exceed overdraft limit");
        }
    }

    void account() {
        System.out.println("accnum :"+accnum);
        System.out.println("balance :"+balance);
        System.out.println("overdraftlimit :"+overdraftlimit);
    }
}

public class Bank {
    public static void main(String []args) {
        SavingsAccount s = new SavingsAccount(2000,2424,5);
        CurrentAccount c = new CurrentAccount(2154,5845,500);

        System.out.println("savings account");
        s.account();

        System.out.println("\ncurrent account");
        c.account();

        System.out.println("\nDeposit");
        s.deposit(1000);

        System.out.println("\nwithdraw");
        c.withdraw(2000);

        System.out.println("\ninterstrate");
        s.calculateintrest();

        System.out.println("\ntransfer");
        s.transfer(c,500);

        System.out.println("\nwithdraw");
        c.withdraw(5000);

        System.out.println("\nfinal account details\n");
        s.account();
        c.account();
    }
}

```

7. Output Screen

```

savings account
accnum :2000
balance :2424.0
interstrate :5.0

current account
accnum :2154
balance :5845.0
overdraftlimit :500.0

Deposit
Deposited: 1000.0

withdraw
Withdrawn: 2000.0

interstrate
value :171.20000000000002

transfer
Transferred: 500.0
To Account: 2154

withdraw
Withdrawn: 5000.0

final account details

accnum :2000
balance :3095.2
interstrate :5.0
accnum :2154
balance :-1655.0
overdraftlimit :500.0

```

8. Result

The Bank Account Management System was successfully implemented in Java using classes, inheritance, constructors and method overriding. Deposit, withdrawal, transfer, interest calculation and overdraft operations were demonstrated successfully.

9. Note

The given program permits the CurrentAccount balance to become negative when the withdrawal is within the overdraft limit. Therefore, the final current-account balance of -1655.0 is expected for the supplied code.